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Handwritten English Script Recognition System Using CNN and LSTM

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***Abstract*— In this digital era, computers and phones are more important than ever, but many people still prefer handwritten documents to digital formats. Sometimes, storing, carrying, and reading through all handwritten documents at once is difficult, so there is a need to convert handwritten documents into digital versions. This paper presents two classification algorithms to convert into digital form: one is classifying words directly using CNN, and the other is character segmentation. The proposed model is based on a convolutional neural network, which was trained using word images from the IAM dataset. The number of layers and dataset size are kept smaller so that training and testing can be done on the CPU instead of the GPU. The proposed model consists of CNN layers for character recognition and LSTM layers for word and syntax correction. The recognition system takes an image as input and displays the digital version of a handwritten document as output after recognition. The CTC loss function and CTC decoder were used after LSTM to align sequences. This paper covers detailed intuition about the architecture and implementation of the proposed system, and its performance was evaluated based on the correct prediction of characters.**

Keywords— Handwritten Script, Digital Text Converter, CNN, LSTM, CTC Loss.

I. INTRODUCTION

People still use so many digital writing tools to write handwritten notes. There are so many drawbacks to handwritten documents. Storing, sharing, carrying, reading, and searching through effectively and efficiently is difficult. Thus, this research paper will help to turn handwritten scripts into text format. Text format will be easy to share, analyze and search through and will help people to access it easily. This model aims to explore the work of classifying handwritten documents and converting them into digital text format, as given in Fig-1.



Fig. 1. Image of the word and its transcription into digital text

Fig-2 shows a handwritten script from IAM dataset. A deep learning model is trained using this dataset, and a RNN-based model is used to correct the words and characters. The dataset has images, so to process these, a CNN is required, which has a number of layers for extracting the features and identification of words and characters [1]. Selecting a feature extraction approach is the most important step towards getting outstanding recognition performance in character recognition systems. In terms of feature extraction methodologies, the invariance features, reconstruct ability, and predicted distortions and variability of the characters are studied. It also emphasizes how challenging it can be to choose the feature extraction technique that is most appropriate for a certain application. Once a few interesting strategies have been identified and evaluated, the optimal feature extraction method for the specific application must be ascertained. Many researchers have done work in this handwritten character recognition field, but mostly with OCR and character segmentation systems, which have low accuracies compared to CNN and RNN-based models [2].

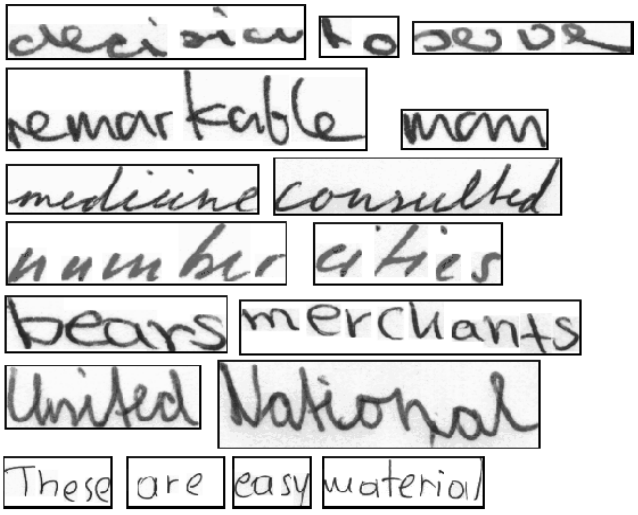


Fig. 2. Handwriting Script from IAM dataset

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When it comes to pre-processing, a ConvNet needs a lot less work than other classification techniques. While filters in primitive methods must be hand-engineered, ConvNets may acquire these properties and filters with enough training. [3]. ConvNets are designed to resemble the structure of the human brain's neuronal connection network, which is modeled after the organization of the visual cortex. Individual neurons can only react when there are changes in the receptive field, a very small visual field area.

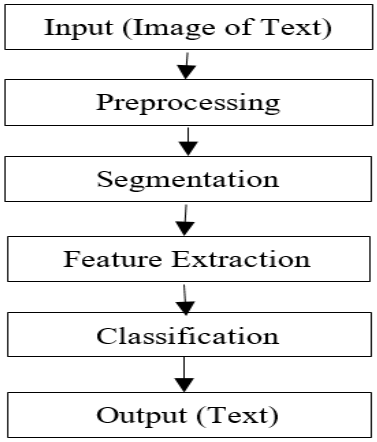


Fig. 3. Feature Extraction Steps for Handwritten Script

Machine Learning requires Early Feature Extraction for features, and classification is conducted on it. Deep Learning, on the other hand, functions as a "black box," extracting and classifying features on its own. The basic job is to classify the supplied image as face or non-facial, as shown in the diagram above. Machine learning requires image attributes such as edges, color, shape, and so on, and classification is done on its own. On the other hand, deep learning extracts features and classifies them on their own. Each layer of CNN takes a feature, and the Fully Connected Layer does categorization [4]. The major conclusion is that Deep Learning uses deep neural networks to extract data and classify itself. Fig-3 shows the feature extraction steps for handwritten script.

Handwritten script to digital text recognition is a very useful system as it is used for various purposes, especially in many real-life scenarios like criminal handwriting detection. This system can recognize and convert the handwriting of a criminal into digital text just by pre-processing and training the image of the handwritten script. This makes this model a very useful and simple tool in many ways. Handwritten document recognition has become an important research area in the field of image processing and pattern recognition in recent years. It involves an automation process that improves the interface between humans and machines. Many research works are focusing on reducing process time and having high recognition accuracy via new methods. Handwriting recognition is a very tough task as people have different handwriting and there are different characters like small letters and capital later, symbols and symbols. The handwritten document is converted to text format either by scanning a handwritten document or by writing with a special pen on the electrical surface. These two methods are, respectively, offline and online processes [5].

Although handwriting recognition on CNN has come a long way in recent years, a significant amount of training data is still required to get respectable results. The number of images can be increased without increasing labor via "data augmentation," or the preparation of massive volumes of image data for training using conventional image processing techniques. However, using traditional data augmentation techniques, it is difficult to address problems associated with a shortage of training data and to produce character representations that seem to have been authored by multiple authors. The proposed method learns from character image data to generate statistical character structure models composed of probability distributions of strokes. By creating strokes based on the probability distribution of each stroke and combining them into a character without altering the source images, character images of different handwriting samples can be produced. In comparison trials with a convolutional neural network for handwritten character recognition, good results were obtained when the proposed method was applied in conjunction with traditional data augmentation procedures.

II. LITERATURE REVIEW

Indiran H.P et al [6] proposed a handwritten character recognition system aiming to classify each handwritten word to convert handwritten text to a digital format. It shows how neural networks can be used to create a system that can recognize handwritten English alphabets. To achieve the machine learning model's desired training, this model uses Python, Machine Learning, ML development IDE software, Computer Vision, CNN, Character Recognition, Classification, and Deep Learning techniques. The National Institute of Standards and Technology (NIST) provided a number of scanned document datasets that were used to build the dataset used in this model. The programme tried to improve by extracting characters from the handwritten image and then classifying each character separately to reconstruct the digital letter based on the final findings with handwritten alphabets. The evaluation parameters used in this model are backpropagation and resilient propagation.

In this model they discovered that, when experimenting with different hyperparameters, keeping the number of epochs low was a good method. This method saves us a lot of time in training, but it also had certain drawbacks. This data augmentation strategy assisted in making the model more resistant to a few tiny but common details that may appear in our test set. They identified the model’s requirements and planned the solution using the knowledge they had described.

Transfiguring Handwritten Text and Typewritten Text was proposed by Keerthana M et al [7] and they developed an application and it can be used to convert handwritten text to editable text or typed text. This model uses Python, PyCharm, GUI, PIL (Python Pillow) softwares and Transfiguring, Randomization, Pattern recognition, Python pillow (PIL), Tesseract, Validation techniques to get the desired prediction for the converted handwritten script. The findings of this research paper show that the output of the converted handwritten script into digital text is shown as a GUI based image on the screens of the user. The model also displays the accuracy of the prediction and the possible error rate for the text. This model uses validation and model accuracy as evaluation parameters to determine the error rate and precision of the model. The major advantage of this model is that the user controls the entire process through a graphical user interface (GUI). Processing and converting both handwritten and typewritten texts take less time and effort with this application. The Disadvantage of this machine learning model

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is that the model focuses less on the accuracy and more on the graphical representation of the output and the text which makes the model less useful for real life purposes and it decreases the experimentation value for the model.

Handwritten Character Recognition Using Deep-Learning was proposed by R. Vaidya et al [8]. This model was trained neural networks in the system using TensorFlow and processed images using OpenCV. This application used the Python programming language to create this system. The software used to develop this model are Python, OpenCV, Android, JAVA, and TensorFlow. This model requires many different types of machine learning techniques to get the desired prediction as output and some of the core techniques used in this machine learning model are Neural networks, Handwriting recognition, Character recognition, Image segmentation, Smartphones, and Text recognition. The datasets used to construct this model were taken from google cloud services and amazon cloud services. Classification accuracy and character recognition accuracy are the two main evaluation parameters for this model and its training dataset. The advantages of this model are: The prediction accuracy of this model is very high and it is 92%, than other two models. This model is also very portable and user friendly to use as it is the form of an android app which is also very accessible. The disadvantages of this model are: This model uses cloud-based services as a dynamic dataset and it might not contain all the required data for handwritten character recognition and pattern recognition which makes the scope of the training model very small and limits the model to recognize only a few types of handwritten scripts.

Y. Singh et al [9] performed a comparative study on various activation function for deep neural networks and suggested best activations according to the application. M. Sohail et al [10] compared various machine learning and deep learning techniques for handwritten digit recognition. Their study results show that CNN based models performed well for handwritten digit recognition. A. K. Bathla et al [11] discussed the challenges in recognition Devanagari script and suggested some solutions. The technique for recognizing handwritten English alphabets and numerals was proposed and developed. It has been experimented with all English alphabets and numerical digits, as well as a variety of handwriting styles. The machine recognized the alphabets and numerals with an average accuracy of 82.5 percent, which is noteworthy and may be acceptable in some applications, according to the findings of the experiments. A. Sharma [12] performed a review for Devanagari script character recognition and described research gaps and findings. M. K. Gupta [13] developed a handwritten OCR system for word in Indic Language. They concluded that machine was shown to be less accurate when classifying similar alphabets, although this misclassification of similar patterns may improve in the future, and a similar experiment can be conducted across a larger data set and with additional optimal network parameters to increase the machine's accuracy.

Sarmah, J et al [14] proposed system for handwritten digit recognition. They used pre-trained model in their system and achieved 98% accuracy. Their results show that LeNet model work well for handwritten digit recognition as this model was designed particularly for this task. M. Lal et al [15] performed a comparative study on pre-trained models for handwritten digit recognition and suggested an efficient model. Mishra, P. et al [16] discussed the extraction of information form OCR documents using neural network.

TABLE 1. FINDINGS OF LITERATURE REVIEW

Contributions / Authors	**Indira, H.P. [6]**	**Keerthana M., Hima Varshini P., Sri Thanvi K., Vijaya G., Deepa V. [7]**	**R. Vaidya, D. Trivedi, S. Satra, and P. M. Pimpale, [8]**
Purpose of study	The purpose of this research work is to classify each individual handwritten word in order to convert handwritten text to a digital format. It shows how neural networks can be used to create a system that can recognize handwritten English alphabets	The primary focus of this application can be used to convert handwritten text to editable text or typed text, as well as to convert typewritten text to handwritten text.	The purpose of this research work and project was to develop a handwritten character recognition system based on image segmentation and deep learning approach using
Technique used	Computer Vision, CNN, Character Recognition, Classification, Deep Learning	Transfiguring, Randomization, Pattern recognition, Python pillow (PIL), Tesseract, Validation	Neural networks, Handwriting recognition, Character recognition, Image segmentation, Smartphones, Text recognition
Findings	Image extraction and handwritten script detection	Detection and conversion of typewritten text	Recognition of handwritten script using deep learning libraries and google services
Dataset Used	NIST dataset	Public databases and old transcripts	Google cloud services and other public databases
Evaluation parameters	Back propagation, Error rate	Validation and pattern recognition	Complexity, Accuracy

A. Ansari et al [17] used a language-based model for text recognition and they used a search technique where a possible final beam search candidate routes is given as penalty/benefit score, as well as their combined individual SoftMax probabilities, which represent the chance of the sequence of characters/words. If the SoftMax layer shows that the most likely candidate word is probably the most likely candidate word, and beam search is quite unlikely given the context so far, the model can correct itself accordingly. Table [1] presents a summary of the research work authored by various researchers and this aims to provide a concise overview of their contributions.

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III. PROPOSED SYSTEM

The proposed work for this research aims to develop a model which uses both machine learning and deep learning approaches to convert the handwritten script by a human to digital text as an output onto the user GUI. Manually transcribing large amounts of handwritten data is a very difficult process that's bound to be fraught with errors and a lot of time. Many applications require the automatic recognition of handwritten text in order to process large amounts of handwritten text. This program is intended only to convert handwritten script to digital text; it is not intended for editing or other forms of manipulation. This model shows how to create a system that can recognize handwritten English alphabets using neural networks.

This task is achieved by using the CNN approach, which uses character segmentation and direct word classification. With regard to the former, a model that can precisely classify words is trained using a variety of CNN architectures. The Real-Time Handwritten Text recognition With TensorFlow and Deep Learning model is a raw python and deep learning-based application which includes a few datasets and libraries to train a model. This Real-Time Handwritten Text recognition With TensorFlow and Deep Learning model will be done using the IAM dataset mainly.



Fig. 4. Text extraction from the IAM dataset

The main purpose of this model is to transcribe and convert the handwritten script contained in scanned images or text files into a digitized form of text. For this application, a training model was added and less error rate was achieved. IAM dataset was used and also (CTC) decoders to integrate them with word beam search. The output for this model will also display the probability of the word being recognized including the word itself.

A. Dataset Description

The publically available IAM dataset from Kaggle was taken for training and testing. 87671 records were taken for training and 9741 records were taken for testing out of total records 97412. This dataset contains the images of handwritten lines of text created by different writers. The dataset is labeled at word, line and sentence levels.

Firstly, extract the (IAM) dataset from its website and download the other required datasets for converting Handwritten scrip to digitized text. The (IAM) dataset contains all the samples of handwritten script to train our machine learning model to recognize and convert the script as shown in the Fig 4.

Secondly, download and import all the required libraries and datasets for training this machine learning model and also create modules for this application. Now create and execute the 6 modules for this application to start predicting and converting the handwritten script into text. A sample pre-processor was built for all the sample stored in the system to be processed. Now load all the data and integrate it by creating a module named (dataloader.py). Now for the user to upload the image of the handwritten script and convert it into the digitized text. A module was created using flask to integrate our machine learning model to a web page, which allow the user to upload the images of their handwritten script and convert them into digitized text. Fig-5 shows the handwritten script recognition system architecture.

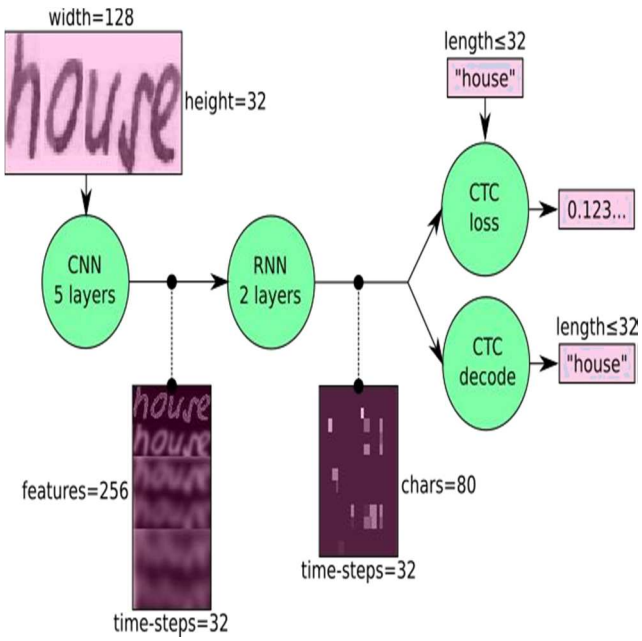


Fig. 5. Handwritten Script Recognition System Architecture

Finally, create the main module to run the application which will execute all the modules and trained model and give us the desired prediction output. The objective of this model is to achieve the final output with the least amount of error rate in the recognition of handwritten script and print the final output as digital text.

B. Model Architecture

The CNN model has one input layer, 11 convolution layer with filter size (3, 3) and (2, 2) and 4 max pooling layers with pooling size (2, 2) and (2, 1). Batch normalization was used and dropout rate was 0.2.

For words and sentences correction bidirectional LSTM was used that has 5 layers. One CTC layer for loss calculation and one output layer. Total trainable parameters are: 26,907,008.

After trained and tested this model is interfaced with web interface using Flask. A handwritten script is given as input and system converts it into digital characters. The efficiency is measured in number of correct prediction of characters.

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This Handwritten Script to Digital Text conversion model can be trained and executed in two ways: The first way to is to train the model with all the samples of the handwritten script and use the (IAM) samples as the main training data. The training model consists of 66 epochs and each iteration of an epoch consists 1258 batches to train and this process of training might take about 7 – 8 hours depending upon the system and the GPU speed. After the training process is complete the trained model gives us an output of the predicted text after converting the saved image of a handwritten script as shown in the Fig 6 below.

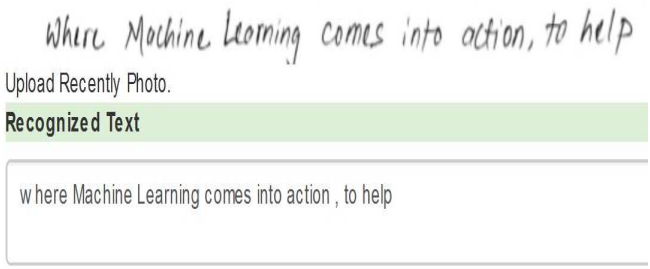


Fig. 6. Predicted output from the trained model

The second method to implement Handwritten Script to Digital Text conversion model the model is to let the user upload an image of their own handwritten script sample onto a web page created using flask and a module named (upload.py). The user is only allowed to upload their sample images in the format of (.png) or (.jpg) onto the web page from their system memory. After uploading the image, the pretrained model predicts the text, converts the handwritten script into digital text, and shows the converted text as output onto the user's screen. The prediction of the converted text has a moderate accuracy and depends on the sample of the handwritten script uploaded, as it converts the image of handwritten script in real – time and very quick.

IV. RESULTS AND DISCUSSION

In this section, the obtained results are presented from the development and implementation of the Handwritten Script to Digital Text conversion model, which utilizes both machine learning and deep learning approaches, with a specific focus on the use of Convolutional Neural Networks (CNN) for handwriting recognition.

A. Dataset Selection and Preprocessing:

In this study, IAM dataset was used, serving as the proposed model's primary training data. This dataset preparation stage is critical in ensuring the model's effectiveness.

B. Model Training and Architectural Choices:

The core of the research approach relies on CNNs, which are leveraged to classify words directly and perform character segmentation. Various CNN architectures are experimented to identify the most suitable model for accurate word recognition.

C. Performance Metrics and Evaluation:

To assess the model efficiency, we used different evaluation parameters, but the most important parameter we considered for this model was the character error rate for the English characters to be recognized and converted in real time.

```
Validation character error rate of saved model: 8.654728%
Python: 3.6.4
Tensorflow: 1.8.0
Init with stored values from ../model/snapshot-24
Without Correction clothed leaf by leaf with the dioappoistmest
With Correction clothed leaf by leaf with the dioappoistmest
```

Fig. 7. Validation character error rate

D. Traning Process:

The training process is computationally intensive, with a model trained over 66 epochs, each consisting of 1258 batches. The time required for training varied based on the system configuration and GPU speed, typically taking about 7-8 hours.

E. Real-Time Handwritten Text Conversion:

The first method we employed involved training the model with a large set of handwritten script samples, primarily from the IAM dataset. Post-training, the model was capable of converting saved images of handwritten script into digital text.

The second method allowed users to upload images of their handwritten script in real time via a web interface created using Flask. The pre-trained model was then responsible for predicting and converting the uploaded handwritten script into digital text. The accuracy of the model in the real-time scenario depends on the quality and clarity of the handwritten script provided by the user. The real-time conversion process is rapid and user-friendly, making it a practical solution for individuals seeking to digitize handwritten content. Fig-7 shows the validation character error in the form of a percentage means how many characters are identified wrongly for given input image script. The web page is deployed using the flask environment to view the result or output of the real-time conversion of handwritten script to digital text on the user interface.

V. CONCLUSION

The digital-to-handwritten script converter is a game-changing link between the conventional and digital information processing domains. This creative tool uses digital technology to improve efficiency and accessibility while preserving the allure of handwritten correspondence. This research focuses on working on and implementing a deep learning model that uses CNN and LSTM. This converter creates a digital version of handwritten scripts that preserves their beauty while making them easy to store and retrieve. It also creates new opportunities for sharing, collaboration, and integration with other digital platforms. After validating the saved model, this trained model has a character error rate of 8.6547%. In order to evaluate various embedding models on a greater variety of datasets, we plan to scale up this research in the future. The future is digital; no one will write by hand or on paper. Everything is done by technology. In the future, people will write on touchpads. In that case, the built-in

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software should be able to recognize the text they write automatically and convert it to digital text, which will make searching and understanding considerably simpler.

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